

Ligjerata 9dh

Energjia kinetike mesatare e molekulave

$$p = \frac{1}{3} n_0 m \overline{v_m^2} \quad / \cdot \frac{2}{2}$$

$$p = \frac{2}{3} n_0 \left(\frac{m \overline{v_m^2}}{2} \right) = \frac{2}{3} n_0 \overline{E_k} \quad / \cdot V_0$$

$$p V_0 = \frac{2}{3} n_0 V_0 \overline{E_k}$$

$$\boxed{n_0 V_0 = N_A}$$

$$p V_0 = \frac{2}{3} N_A \overline{E_k} = R T$$

$$\overline{E_k} = \frac{m \overline{v_m^2}}{2} = \frac{3}{2} \frac{R}{N_A} \cdot T$$

$$k = \frac{R}{N_A} \Rightarrow \text{konst. e Boltzmannit}$$

$$\boxed{\overline{E_k} = \frac{3}{2} k T} \quad \text{ose} \quad \boxed{E_{tr} = \frac{3}{2} k T}$$

$$p = \frac{2}{3} n_0 \overline{E_k}$$

$$n_0 = \frac{3p}{2 \overline{E_k}} = \frac{3p}{2 \cdot \frac{3}{2} k T} = \frac{p}{k T}$$

$$n_0 = \frac{p}{k T}$$

Shpejtësia kuadratike mesatare e molekulave

$$\overline{v_m^2} = \langle v^2 \rangle$$

$$\overline{E_{tr}} = \frac{3}{2} k T$$

$$\frac{1}{2} m \overline{v_m^2} = \frac{3}{2} k T \quad / : m$$

$$\overline{v_m^2} = \frac{3 k T}{m} \quad / \sqrt{\quad} \Rightarrow \sqrt{\overline{v_m^2}} = \sqrt{\frac{3 k T}{m}}$$

$$k = \frac{R}{N_A}$$

$$\overline{v_m^2} = \frac{3 R T}{m \underline{N_A}}$$

$$M = \frac{m}{n} \quad n = \frac{m}{M} = \frac{N}{N_A}$$

$$\boxed{N=1}$$

$$\underline{N_A} = \frac{M}{m}$$

$$\overline{v_m^2} = \frac{3 R T}{M} \quad / \sqrt{\quad} \Rightarrow \sqrt{\overline{v_m^2}} = \sqrt{\frac{3 R T}{M}}$$